Cover Letter — Inhyeok Hwang

XAI · MECHANICAL ENGINEERING TUTOR (REMOTE)

Dear xAI Hiring Manager,

The first time I won something for following a question to its physical answer, I was in middle school. I had noticed that spinning a plastic bottle while draining it emptied faster than holding it still — and instead of leaving it there, I traced it: rotation generates a vortex that pulls air in through the center while water exits around the outside, eliminating the pressure lock that stalls a static drain. First place in a school science competition. The habit was set.

I came to mechanical engineering for a practical reason: Korea's manufacturing economy makes it a reliable career path. What I did not expect was that it would give me a way of seeing. The full weight of that arrived years later, at a startup building an agricultural robot — moving between motor control, tracked-vehicle dynamics, CAN communication, GPS integration, and waterproofing failures, sometimes in the same week. The common thread was always physics. Software is governed by hardware; software's job is to extract the maximum performance from what hardware can physically deliver. That lens is what eventually led me to graduate research, and what brought me to xAI's Mechanical Engineering Tutor role.

At Konkuk University's RBDO Lab, I spent two years acquiring failure physics across three structurally distinct material systems — IGBT bond-wire lift-off, photovoltaic polymer degradation, and IPMSM drivetrain fault propagation. For each: unit analysis first, governing model second, test to challenge it third. On the IPMSM work, I implemented fault modes in Ansys Maxwell FEM, co-simulated with a MATLAB/Simulink inverter model, and validated against experimental data — a full pipeline from fault physics to Sim2Real verification.

At GINT Corp., the same discipline applies under a harder constraint: production reality. The pattern is identify the mechanism, find the realistic compromise that can be closed now, solve it — then reflect the root cause through DFMEA and PFMEA so the next-generation design eliminates the failure mode structurally. Four root-cause analyses closed to verified corrective action. Five high-priority DFMEA items resolved before production ramp. Twenty-seven nonconformance reports tracked through closure.

That discipline — mechanism first, assumptions named, limits of applicability stated — is what distinguishes useful AI training data from noise. My three publications span three domains because I spent two and a half years acquiring each domain's failure physics from primary literature. That breadth is evidence that this is a practice, not a one-time result. When xAI lists autonomous learning and technical writing as preferred qualifications, that is a direct description of how I have worked since a middle school science fair taught me to follow a question all the way down.

The remote, part-time structure fits alongside my current industrial role — I can contribute immediately. I would welcome the opportunity to discuss how my publication record, cross-domain ME depth, and physics-first practice can add value to xAI's training data quality.

Sincerely, **In-hyeok Hwang** dlsgur5560@gmail.com github.com/hwanginhyeok hwanginhyeok.github.io/portfolio